

EID 103 Final Report: Mechanical Rectifier

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Project Description:

The purpose of a mechanical rectifier is to take a non-constant input and convert it into a constant output. In this case, the input was non-constant rotary motion, where power was input as a sinusoidal function. There were several constraints the mechanical rectifier had to follow. For one, the assembly had to be quick to assemble. Specifically, it had to be possible to assemble the mechanical rectifier in less than 5 minutes. In addition, the system had to have a feature to mesh with the encoder gear, as well as measure the input and output rotational speeds. The output rotation had to spin a disk within 25.4 mm(1 inch) and 76.2 mm(3 inches). Furthermore, the system could not simply be compatible with one frequency: It had to work for frequencies of 1Hz, 0.5Hz, and 0.25Hz. The system also had to take less than 10 seconds to reach a constant state. Finally, the design of the system had to be safe: No parts could spin more than they were meant to, as that would pose a considerable safety hazard.

Engineering Design Process & Iterative design

The initial objective was to research a multitude of new systems, mainly through mapping out systems in Plectica, and to create a reliable format for the prototype. Ranging from ratchets to rubber gears and bicycle spokes, the team's research process develops not only on the system's desired dimensions, but also on what specific parts may be used for the mechanical rectifier. When the drafting process started, multiple sketches were made as a list of potential machine concepts, specifically narrowing on four to decide on, measuring out the main pros and cons, in terms of efficiency and capability to manufacture and assemble.

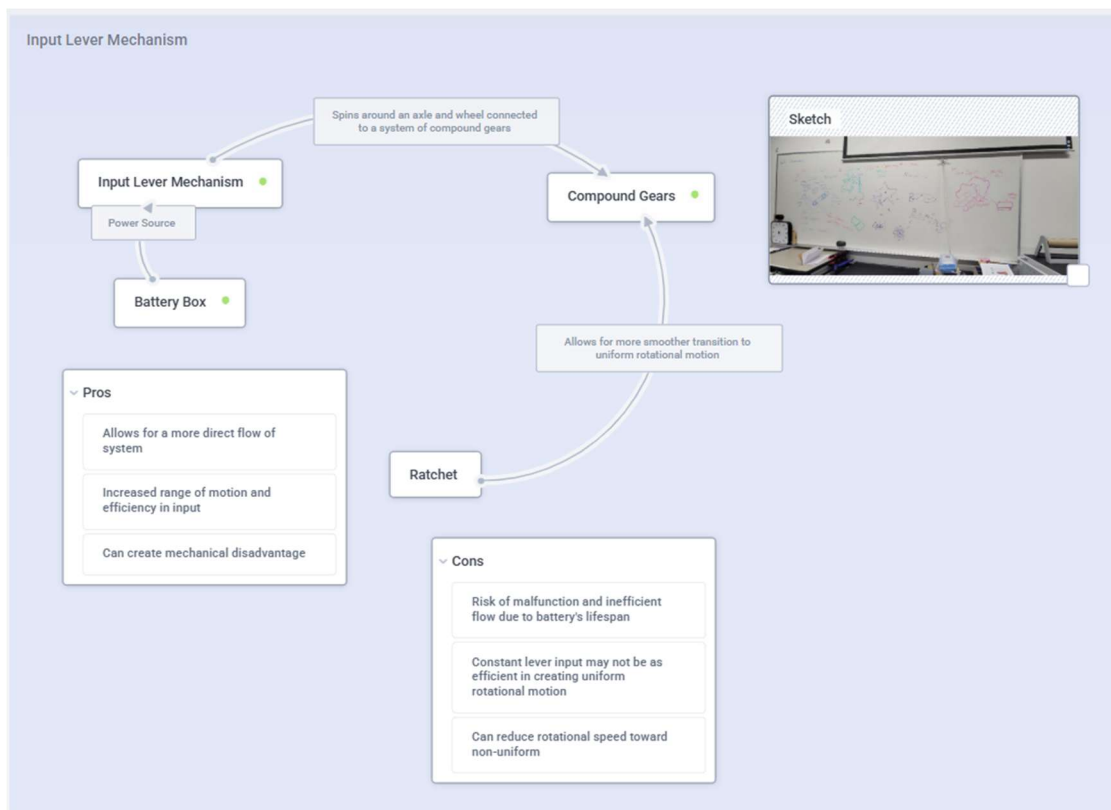


Figure 1: Input Lever Mechanism

The proposal for the input lever mechanism, designed to balance out the non-constant motion of the gears through ratchets. While this can help raise efficiency in input, alongside having a greater mechanical disadvantage and flow of system, this can lead to risk in terms of battery, alongside possibly even reducing the rotational speed.

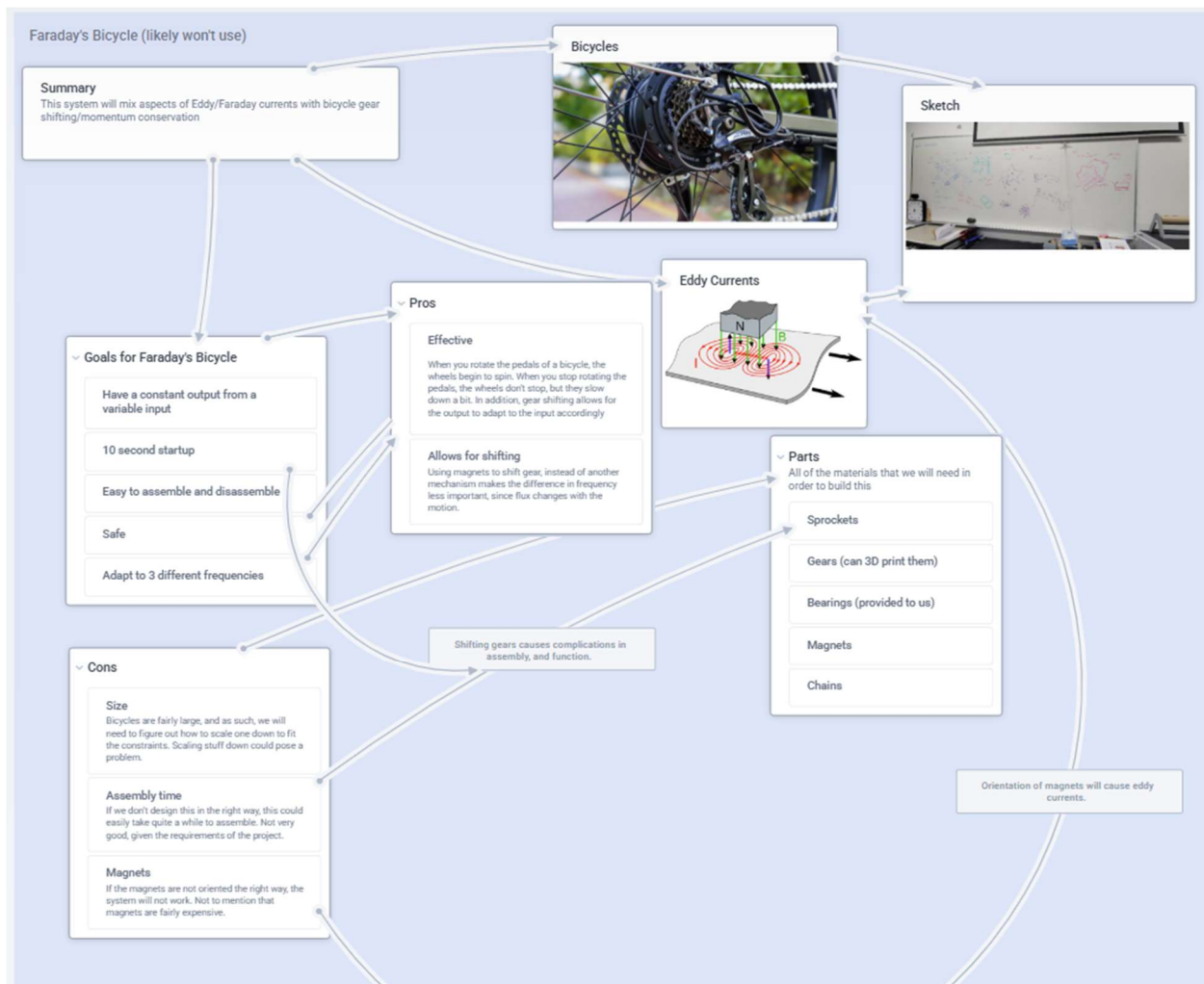


Figure 2: Faraday's Bicycle

The proposal for a design utilizing Faraday's bicycle alongside special magnets and chains for supports, where the magnets' orientation will bring Eddy Currents to create lesser differences

in frequency. While this can be easy to assemble, and adaptable to other variable rotational speeds, the project is also time-costing, alongside having complications in assembling and functionality.



Figure 3: Hydraulic Centrifugal Governor

The proposal for the hydraulic centrifugal governor, utilizing governor circles and new concepts of hydraulic mechanics, through its output mechanism of a fluid container, connected to a petal gear. While its system can provide usefulness towards whole, unique machines, alongside having simple travel within its subsystems, this inclusion of hydraulic mechanics may be complicated, inefficient, and possibly dangerous, considering the density of common liquids, as with water.

Centrifugal Governor Supports

For the centrifugal governor to stand on the breadboard while rotating, supports must be present to hold it in place. The support base acts as the first support, preventing motion at the bottom of the rod containing the centrifugal governor. It is designed to fit a bearing as well as the Long 8mm Rod, allowing rotational motion.

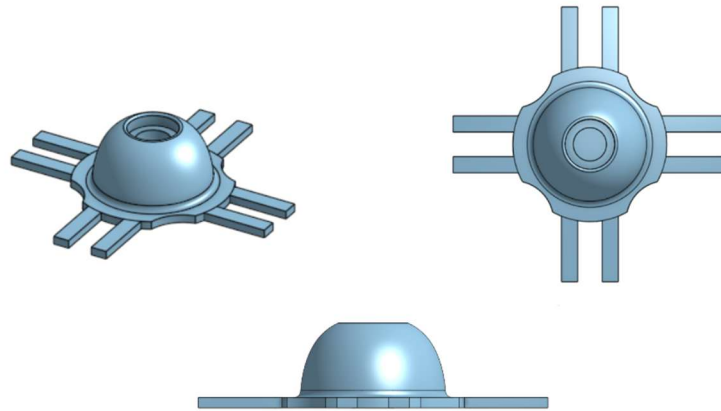


Figure 4: Large 8mm Rod support with top down view and side view.

After the first round of testing, the group determined that another support would be necessary, as the centrifugal governor was unstable even with the support base. This is because the support base only acted at the bottom of the rod: There was nothing holding the rod up closer to the center of mass.

As such, a modular table was made to support the rod further up. It consists of 4 legs screwed into the breadboard and a tabletop with a hole for the rod to go through. The legs are long enough so that the table provides support above the bevel gears, allowing for the bevel gears to rotate without interference.

The modular table is shown below:

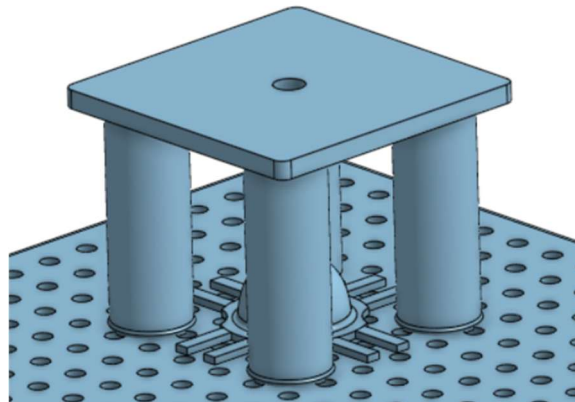


Fig 8. An assembly showing how the table and support base fit together on the breadboard.

Power Transfer System:

In the team's first model, the team decided to create a pulley system. To mesh the pulley to custom gear, the pulley had three-prongs that fit into the custom gear. While the concept worked well, the prongs were too unstable, due to not being filleted at the surface. Additionally, the rubber bands that converted the motion of the Three-pronged pulley to the other pulley would constantly slip due to the smooth concavity of the pulley depicted below:

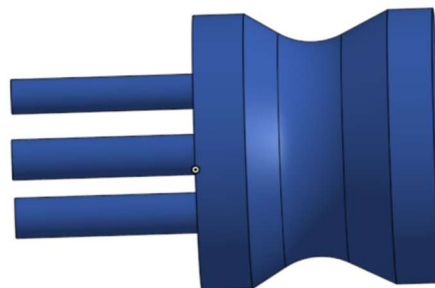


Fig 9A. The pulley that is attached to the encoder to hold the rubber band pulley system.

In Model 2.0, the prongs were filleted at the surface of the pulley. Additionally, the rubber band slippage was resolved using a sharper geometry based on wells, rather than smooth concavity.

This design is shown below:

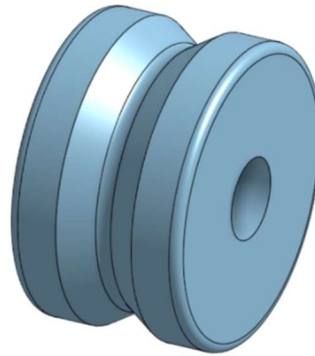


Fig 9B. The final design of the pulley.

Centrifugal Governor Spline:

In Model 1.0, an initial spline was to be created to allow the centrifugal governor to move linearly up and down the Long 8mm Rod. The handles below could not be 3D printed properly due to thin walls and the lack of precision within the 3D-Printer. The team decided that to allow the spline to translate smoothly, the inner diameter of the spline had to coincide with the diameter of a linear bearing(8.2mm).



Fig 10. The initial design for the governor spline.

In the final model of the spline, the team decided to decrease the diameter of the spline and use 3D printer's mid-tolerancing instead of using a linear bearing. This choice was made as the weight of the linear bearing impeded the centrifugal effect of the weights attached to the centrifugal governor. Additionally, a crevice was added so a rubber bands would connect firmly to the spline, allowing transfer of rotation to the next subsystem.

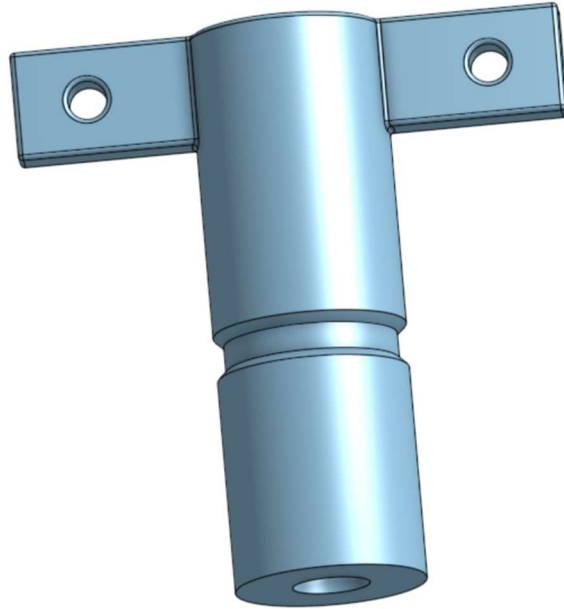


Fig 11. The final design for the centrifugal governor spline.

Bevel Gears:

In Model 2.0, the team would, at first, utilize bevel gears attached to two rods, one horizontally and vertically aligned, allowed to rotate with a clamp. However, due to size difficulties, rather it being too small for the rod to have a friction fit, the team has decided to remove the clamp altogether, instead focusing on only the poles for supports.

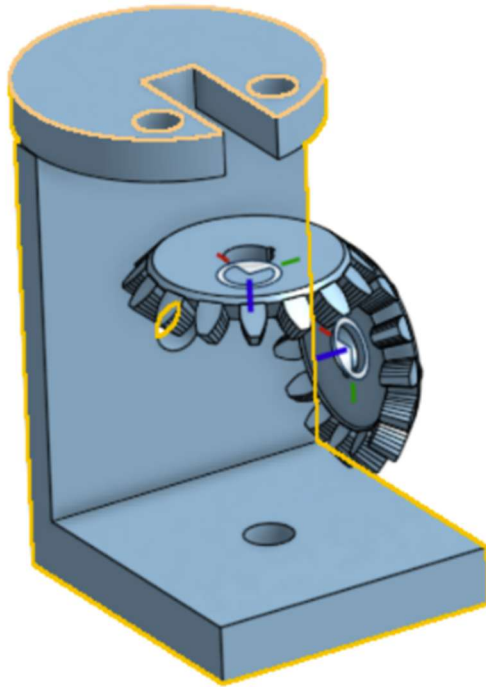


Fig 12. The original bevel gear design.

Cup, Table + Output Gear:

In Model 1.0, it was initially planned to create a large cup filled with water and a gear strongly attached to the bottom through a makeshift hole. However, due to no agreement in dimension between the two, alongside initial confusion upon workspace units, printing it would lead to incompatible parts that would cover the breadboard's majority. Therefore, the group decided to create a flat platform for the cup.

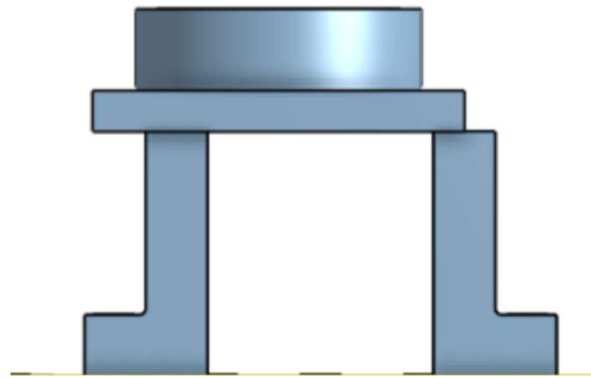


Fig 13. An assembly showing the first version of the cup base.

After the group revamped the entire design of the mechanical rectifier, it was decided that the cup base did not need to be elevated, since the final output would be located next to the cup rather than under the cup. Hence, the cup base would be screwed directly onto the breadboard.

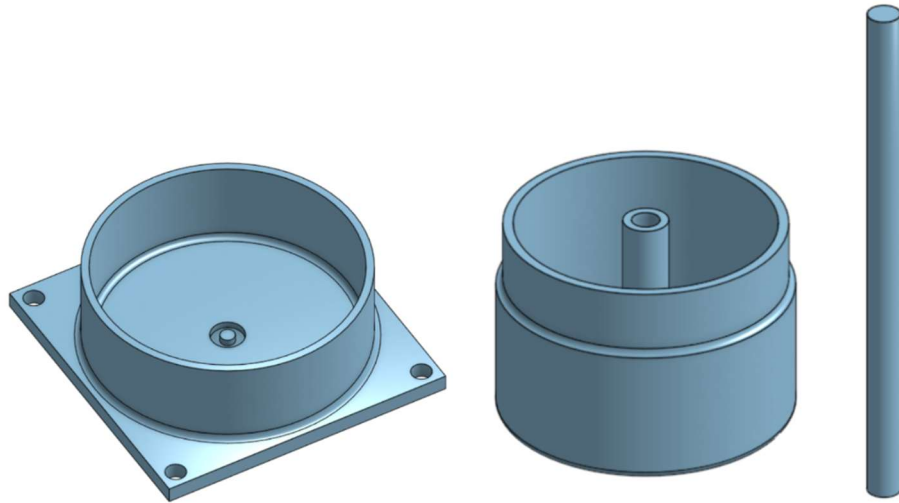


Fig 14. The final versions of the cup base, cup, and the rod for the cup.

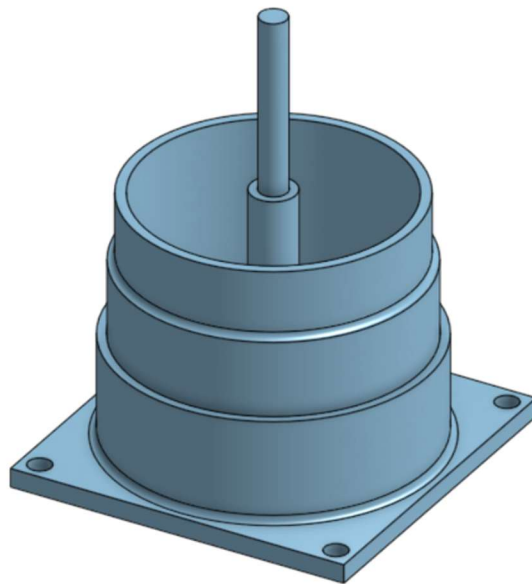


Fig 15. An assembly showing how the base-cup-rod system fits together.

Propeller Spline:

In the first design for the propeller spline, the propeller petals were directly modelled onto the spline. Due to 3D printing limitations, only one layer of petals could be on this spline.

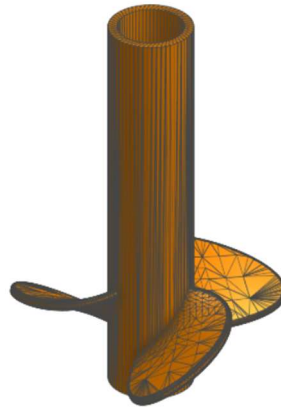


Fig 16. The initial design for the petal spline.

After deliberation, however, the group determined that the spline needed at least 2 layers of petals on the spline. This is because the design operates on the difference in the amount of petals submerged in the cup when the centrifugal governor arms are at different heights: When the arms are higher, less of the petals will be submerged, and when the arms are lower more petals will be submerged. At any given time, petals must be submerged in the water to ensure cup rotation.

To combat this issue, the propeller spline is broken up into 2 components: The spline itself, and a clamp containing petals. The spline is designed to be long enough to hold multiple petal clamps, allowing for multiple layers of petals. In addition, near the top of the spline there is an indent that allows the rubber band to stay steady.

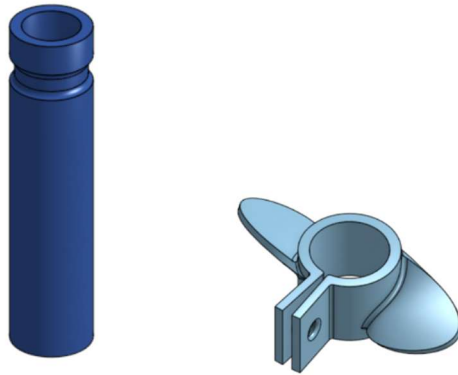


Fig 17. The final designs for the Large Spline and petal clamp

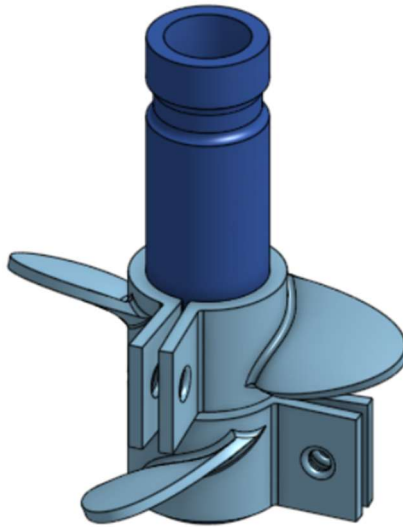
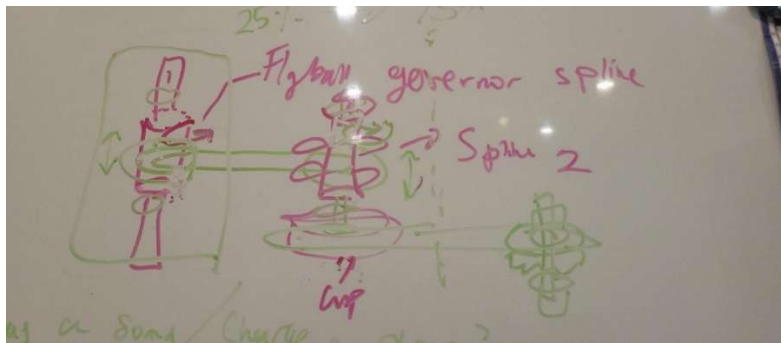
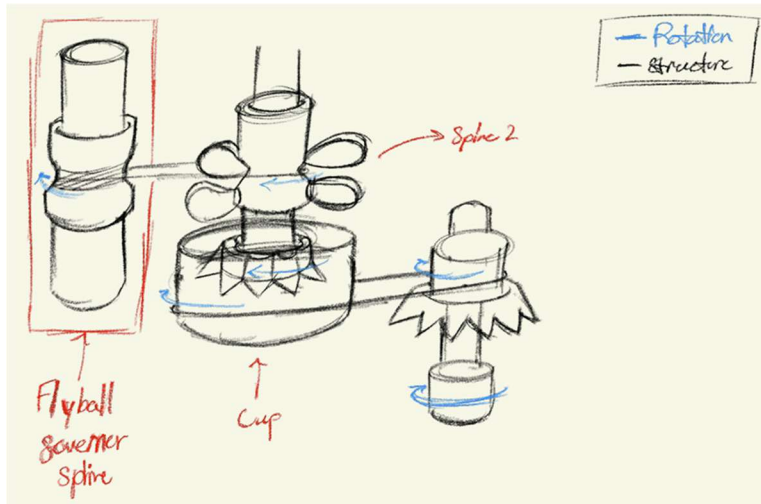


Fig 18. An assembly showing how the Large spline shaft and clamp fit together. There are two layers of metals instead of one layer like in the original design.

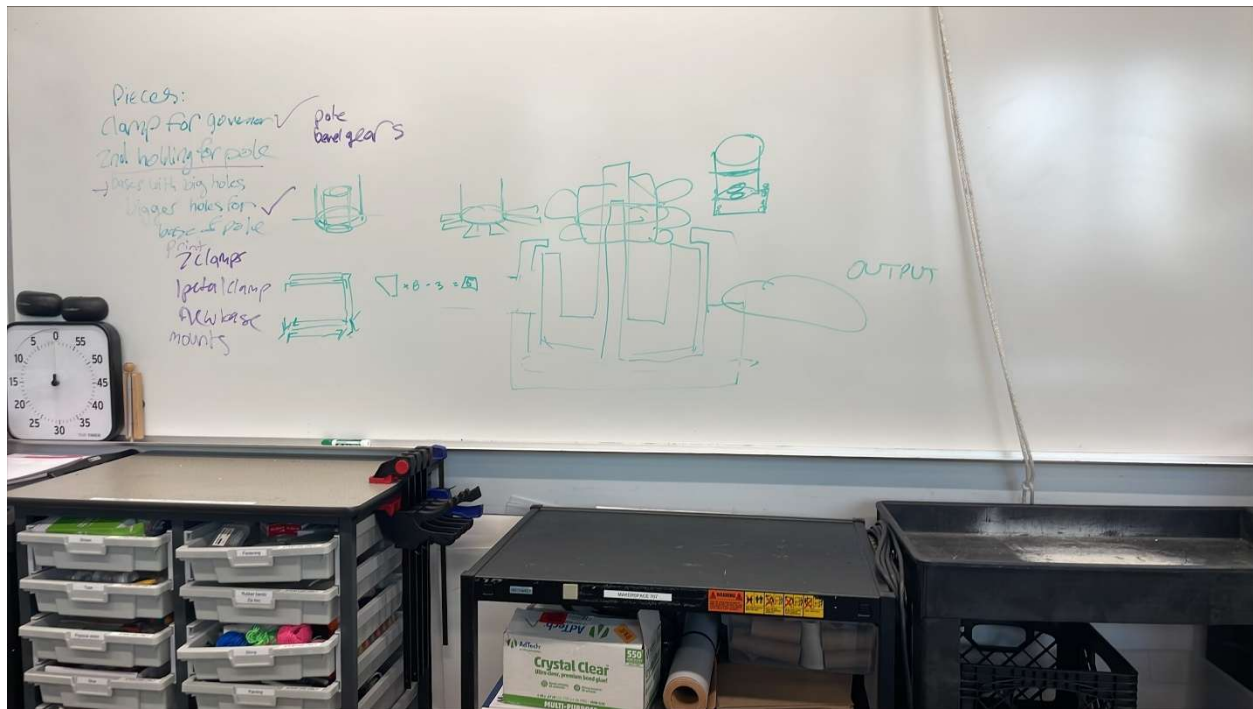
The iterations above resulted from the following:

In the middle of the project the group experienced a large problem. No progress was being made to create anything physical. Many designs and cads were created but issues of dimensioning and how parts would fit together arose and the group was not able to resolve all of these issues in order to print out the project and test it. Since they were unable to test if they did not know how if the project was going to work and what big changes would need to be made. They decided to take a moment to rethink the project. They realized the problem was that they were trying to stack all the subsystems on top of each other and this was causing issues with fitting and supporting the system. They decided to get a working centrifugal governor to print so they could test it and then figure out what to do next. They re-examined all their designs and created a new design that would separate all their subsystems onto the breadboard. This new design effectively resolved their issues with supporting the system and to better visualize the power transfer. Furthermore, their completed centrifugal governor allowed them to design the other subsystems to fit around it.

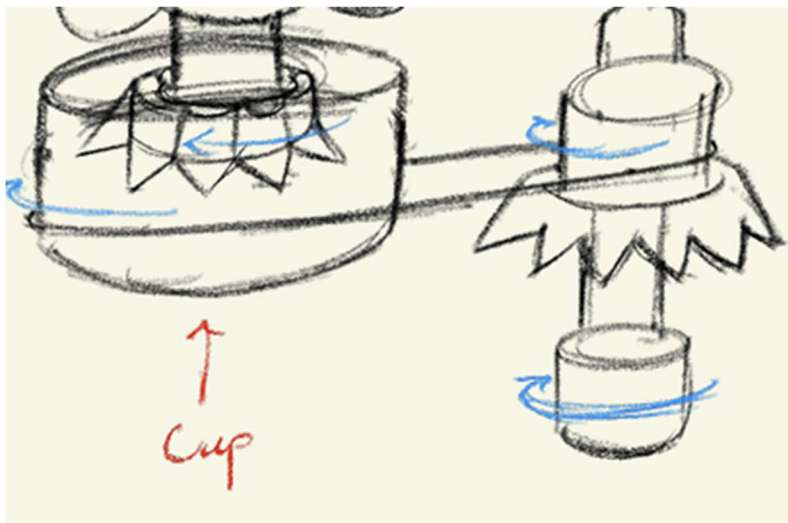
The images below are sketches of the team's final design, after the team brainstormed about the above iterations and redesigning the system such that complicated structures did not have to be involved.

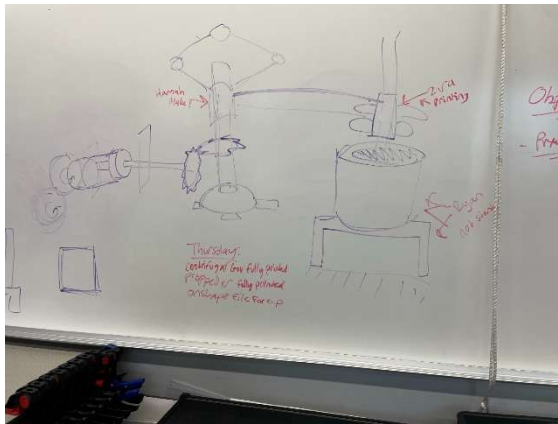
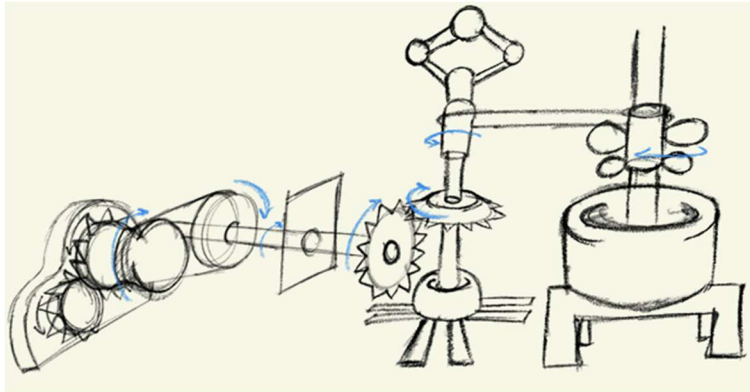
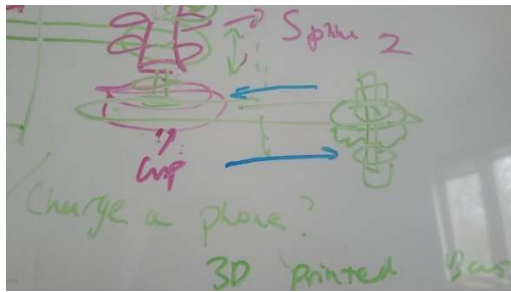


Figs 19 + 20. This image shows one of the many redesigns of the prototype, involving bevel gears to rotate the gear's direction and fluid forces to stabilize the system, emphasizing on the cup and propeller's significance.



Figs 21 + 22. This image depicts sketches of individual parts of the final design, with a heavy emphasis on the cup subsystem.

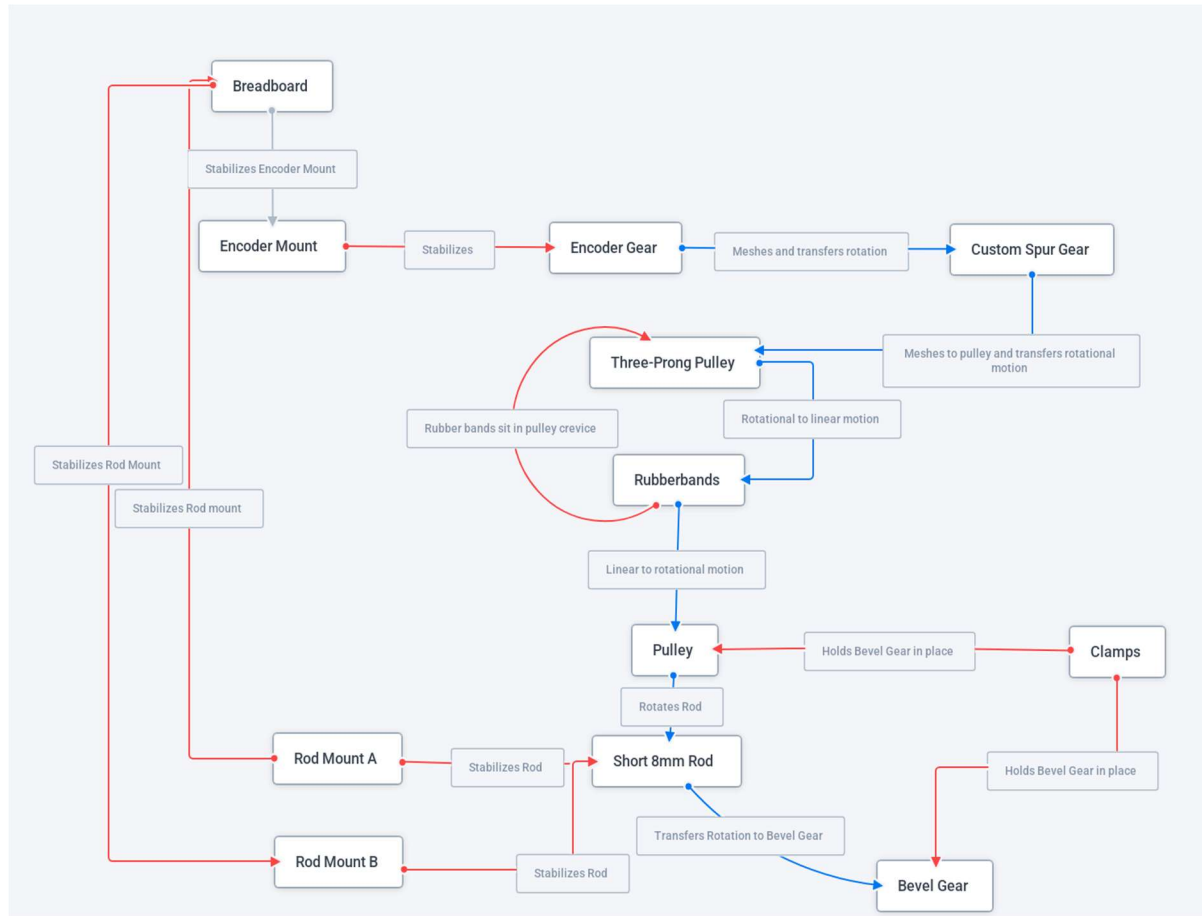




Figs 23 + 24 + 25. These are closer views of the sketch of the final design. This view emphasizes the connection between the cup subsystem and the output.

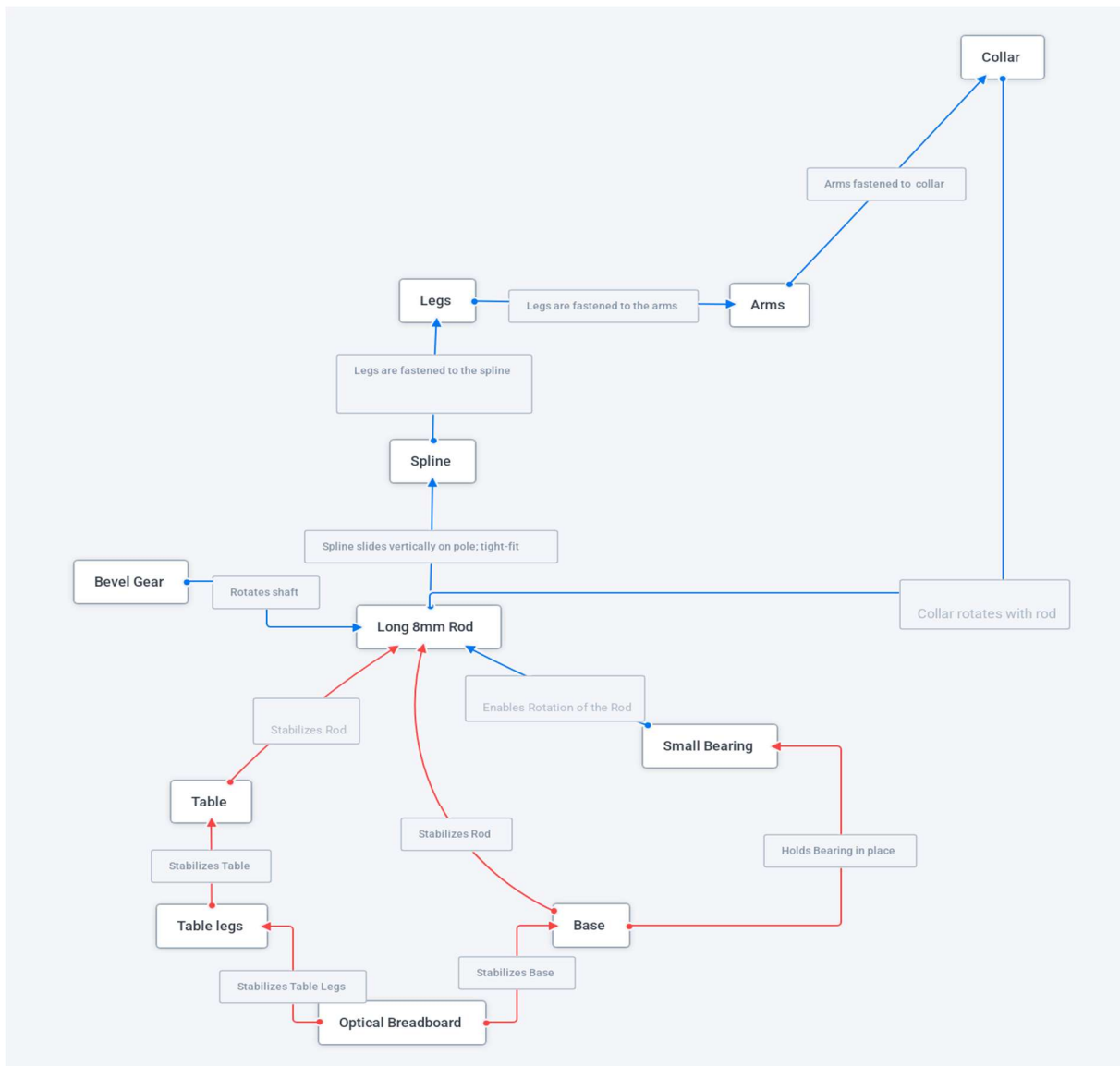
Analysis of the design:

Power Transfer:



To transfer the non-uniform motion to the system, the team decided on a pulley system. As the Encoder gear rotates, the three-pronged pulley rotates. This motion is transferred from the three-pronged pulley to a second pulley, which is attached to a Short Rod 8mm in diameter. Attached to the end of the rod is a bevel gear. As the rod rotates, the bevel gear rotates at the same rate, transferring motion by 90 degrees to the next subsystem.

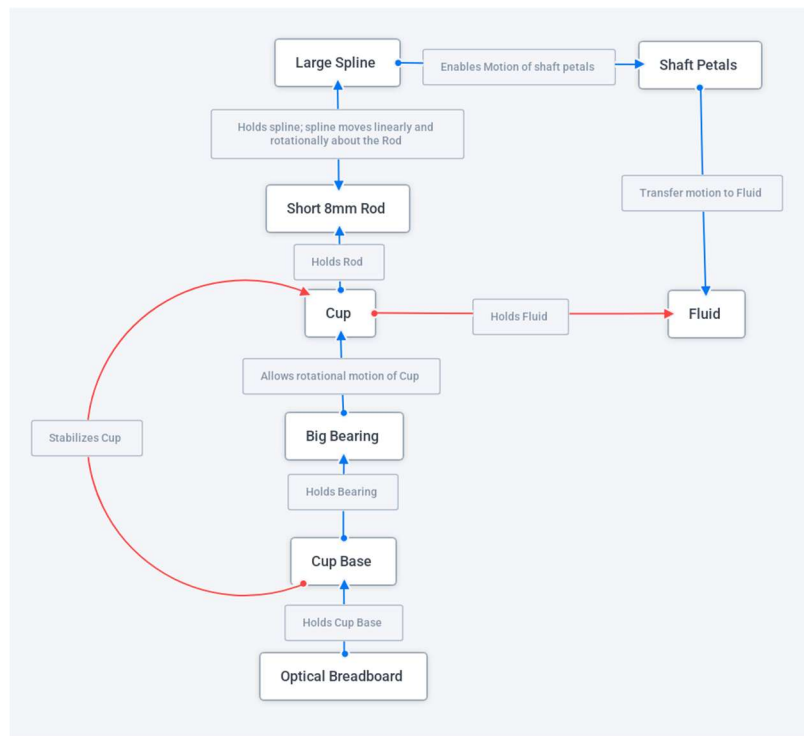
Centrifugal Governor:



The Centrifugal Governor arms hold weights and are attached to the Large 8mm Rod through the collar. The Large 8mm Rod rotates due to a bevel gear meshing to the previous subsystem. As the Large 8mm Rod rotates at a faster rate, the centrifugal force(force that causes something to spin in a circle) produced due to the mass of the weights, causes the arms to rise.

Since the motion from the encoder is non-uniform, the arms of the centrifugal governor effectively buff the change in rotational velocity. This vertical movement is captured through the legs and Small Spline of the centrifugal governor. Since the legs are connected to both the arms and the Small Spline, the vertical movement of the arms is transferred to the spline. As a result of this, the spline has linear and rotational motion along the Long 8mm Rod that it can transfer to the Fluid subsystem through rubber bands.

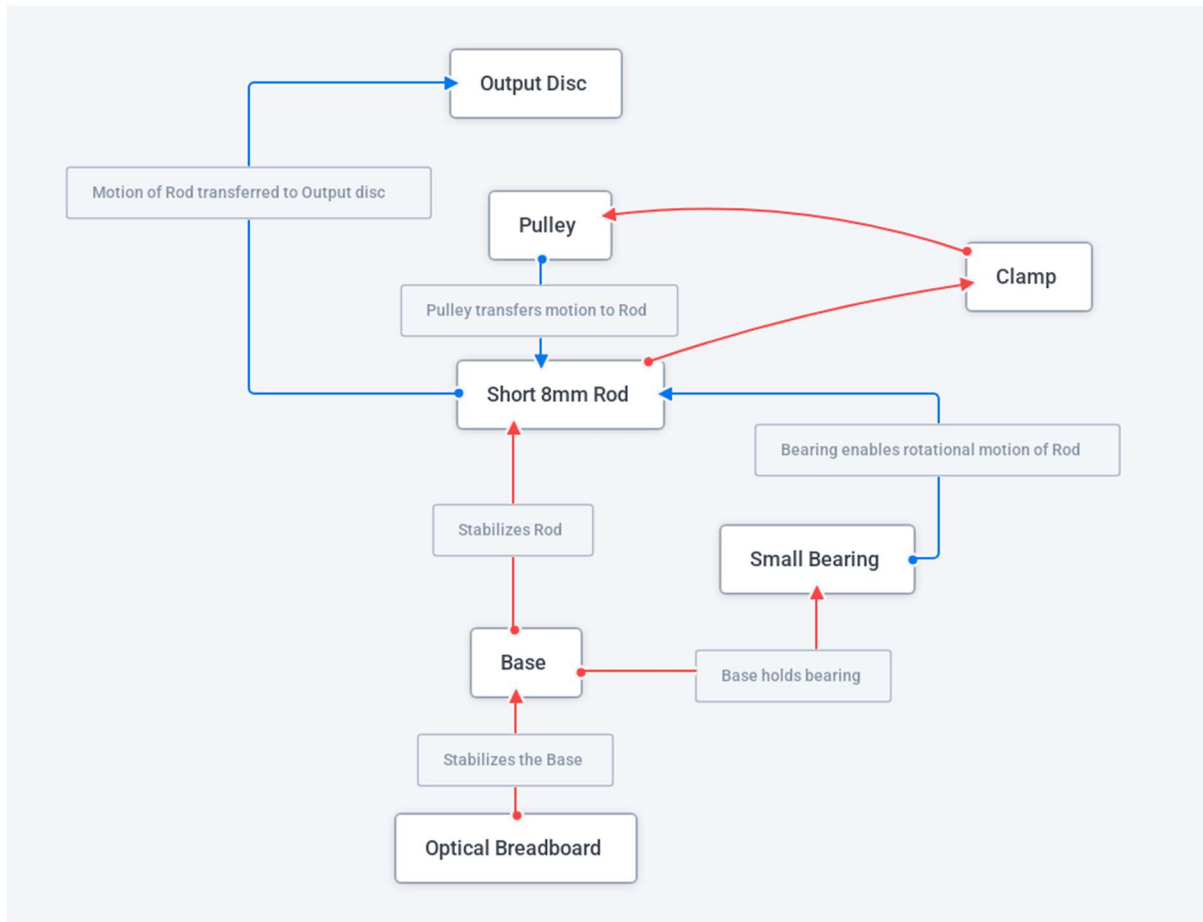
Fluid Subsystem:



Attached to the cup through a Short 8mm Rod is a Large Spline. Due to the spline in the previous subsystem, the Large Spline gains linear and rotational motion. Attached to the Large Spline are Petals which follow the same motion. The Petals convert rotational motion to the fluid. The momentum of the rotating fluid causes the Cup to rotate. Referring to previous subsystem

analysis, as the non-uniform motion changes, the number of petals in the fluid changes. Consequently, the rotational speed of the fluid and the cup change to account for the non-uniform motion. Connected to the Cup are rubber bands, which transfer the rotational motion of the Cup to the next subsystem.

Output:



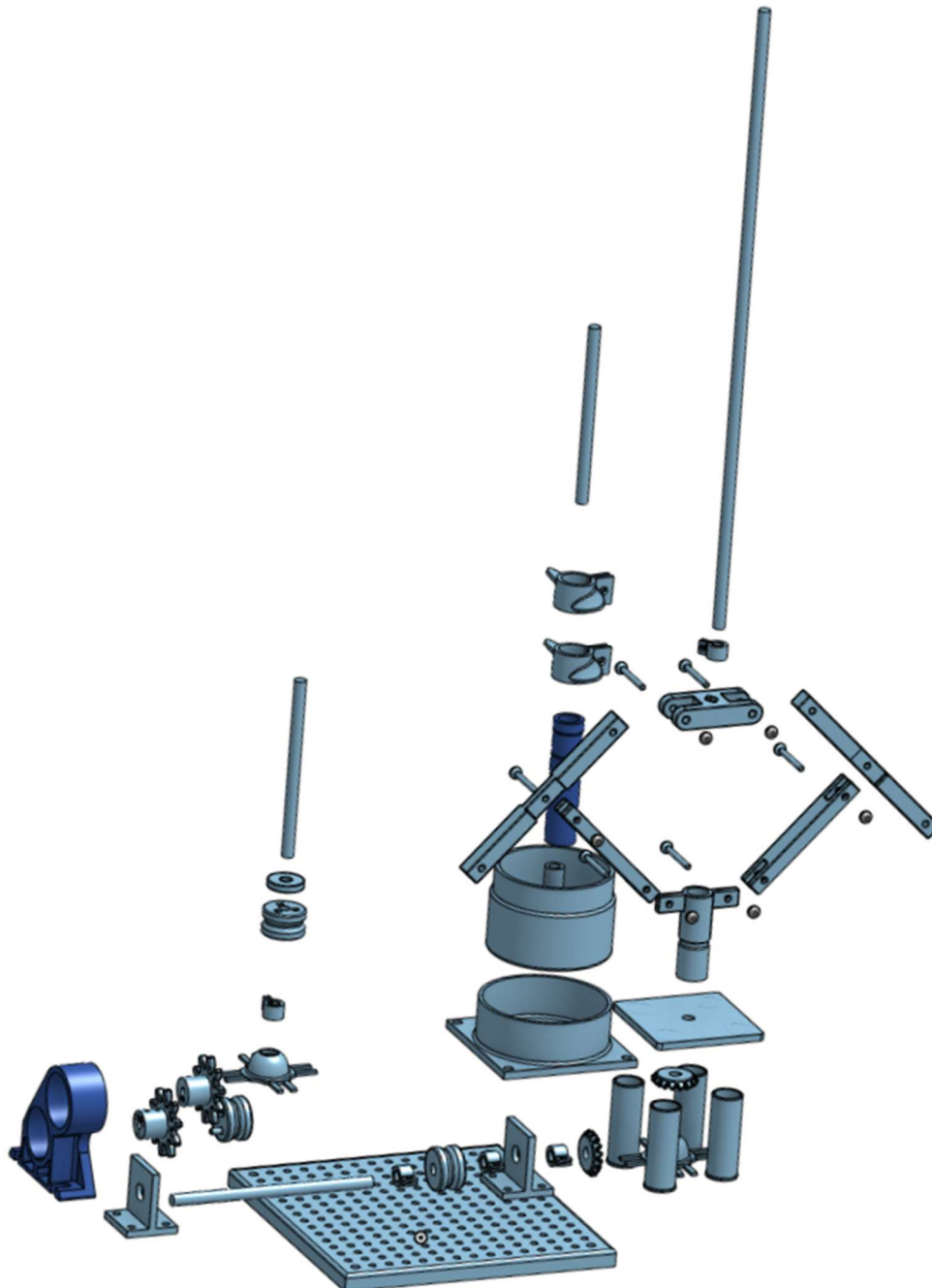
To transfer the rotational motion of the previous subsystem to the Output, rubberbands are placed on a pulley, connecting the Cup of the Fluid Subsystem to this subsystem. The pulley is

connected to a Short 8mm Rod to enable rotation. An output disc is attached to this rod to observe the final output of the Mechanical Rectifier.

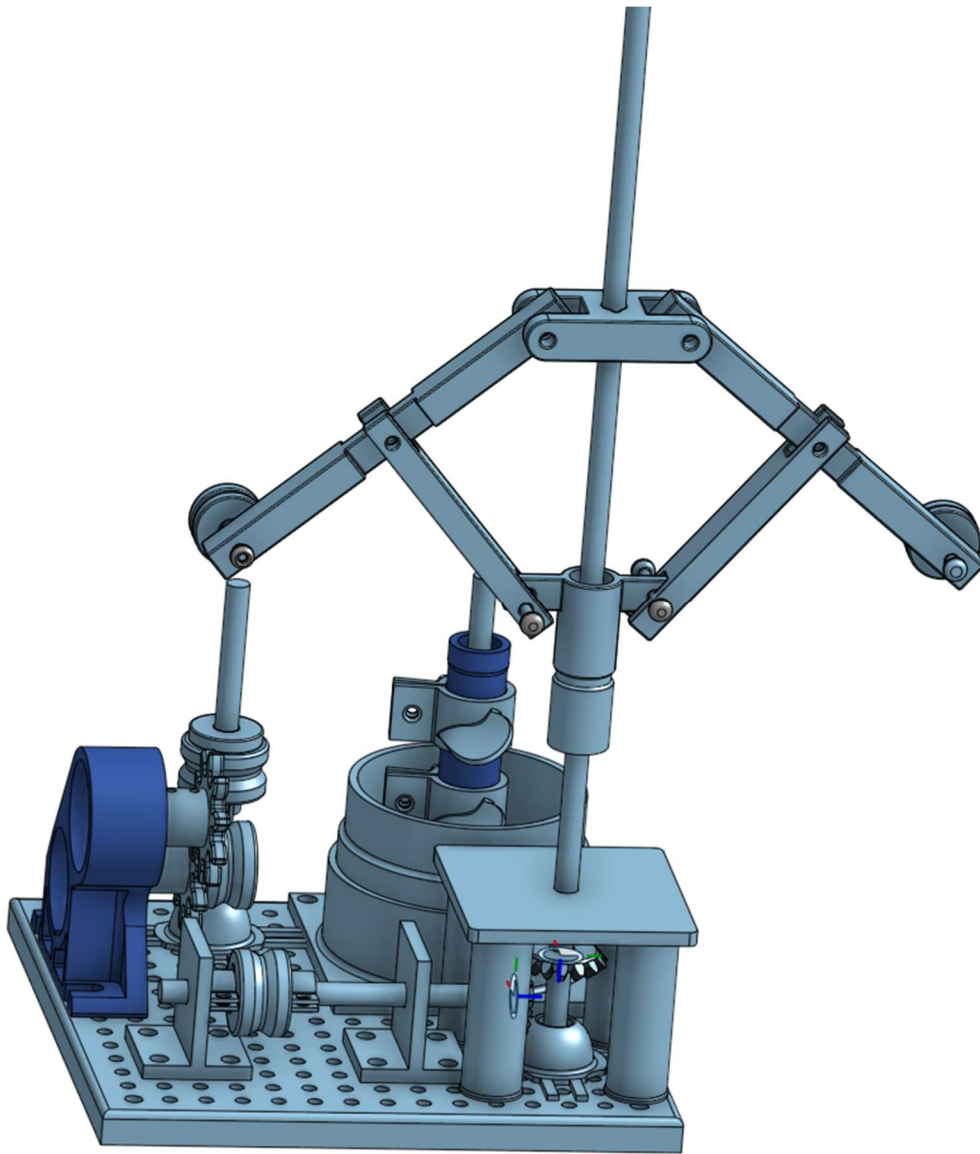
Bill of Materials:

Component name	Quantity	Manufactured/Source	Cost	Brief Description
Arms	2	3D Print	N/A	Connect the centrifugal governor to the Small Spline
Base	2	3D Print	N/A	Stabilizes vertically placed rod components
Bearing big	3	Supplier(Cooper Union)	N/A	Enables rotation of Short 8mm Rod and Long 8mm Rod
Bearing small	1	Supplier(Cooper Union)	N/A	Enables rotation of Cup
Bevel Gear	2	3D Print	N/A	Transmits motion 90 degrees
Collar	1	3D Print	N/A	Attaches centrifugal governor to Long 8mm Rod
Cup	1	3D Print	N/A	Holds fluid and acts as a pulley
Cup Base	1	3D Print	N/A	Stabilizes the Cup
Custom Spur gear	1	3D Print	N/A	Meshes to the encoder gear
Fastener Pin	2	3D Print	N/A	Fastens weights onto Legs
Large Spline	1	3D Print	N/A	Enables rotational and linear motion of Shaft Petals
Legs	2	3D Print	N/A	Holds weights on ends; functions by centrifugal forces
Long 8mm Rod	1	Supplier(Cooper Union)	N/A	Enables motion
M3 Nuts	17	Supplier(Cooper Union)	N/A	Secures arms to legs, table legs to optical breadboard, arms to collar
M3 Screws	17	3D Print	N/A	Secures arms to legs, table legs to optical breadboard, arms to collar
M5 Screws	16	Supplier(Cooper Union)	N/A	Secures components to optical breadboard
Output disc	1	3D Print	N/A	Used to evaluate output motion
Pin Lock	2	3D Print	N/A	Secures Fastener Pins
Pulley	3	3D Print	N/A	Transfers rotational Power
Retaining Ring	5	3D Print	N/A	Secures bevel gear and pulley in power subsystem, pulley in output subsystem and
Rod Mount	2	3D Print	N/A	Secures Short 8mm Rod in Power Transfer subsystem
Shaft petals	2	3D Print	N/A	Cause fluid to rotate
Short 8mm Rod	3	Supplier(Cooper Union)	N/A	Enables motion of Power Transfer Subsystem and Output Subsystem
Small Spline	1	3D Print	N/A	Enables linear and rotational motion of the Fluid Subsystem
Table	1	3D Print	N/A	Stabilizes the Centrifugal Governor Subsystem
Table leg	4	3D Print	N/A	Stabilizes and attaches Table to Optical Breadboard
Three-Prong Pulley	1	3D Print	N/A	Meshes to the Custom Spur gear to transfer Encoder Gear motion
Washers	8	Supplier(Cooper Union)	N/A	Add weight to the Centrifugal Governor
RubberBand	9	Supplier(Cooper Union)	N/A	Connect pulleys of the Power Transfer Subsystem, Small Spline to Cup, Cup to Pulley in Output Subsystem
Optical Breadboard	1	Supplier(Cooper Union)	N/A	Acts as a base and stabilization for the whole system
Encoder Gear	1	Supplier(Cooper Union)	N/A	Provides non-uniform motion to the system

Exploded view of assembly:



Normal View of Assembly:



Outcome Analysis:

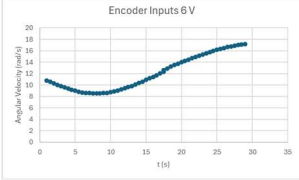
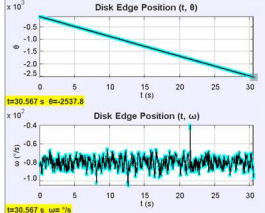
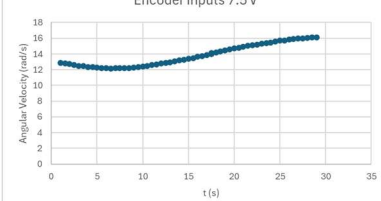
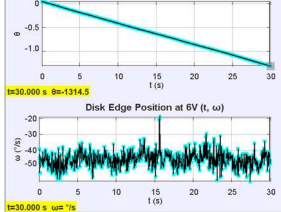
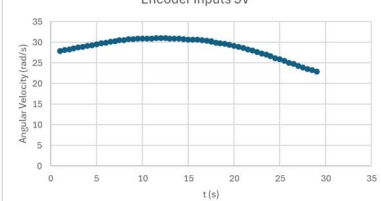
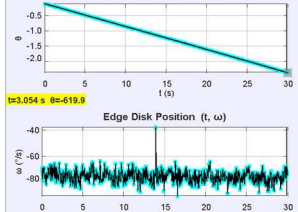
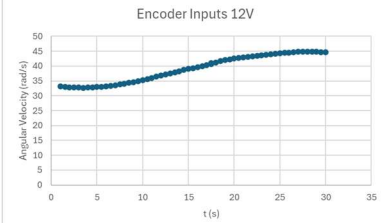
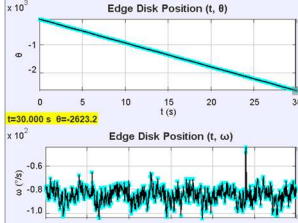
Voltage	Encoder angular velocity(radians/second)	Rectifier angular velocity(degrees/second)
6V		
7.5V		
9V		
12V		

Table 1: Graphs of Angular velocity vs time for non rectified and Mechanical rectified system for the four tested voltages and Graphs of the Mechanical Rectifier system of angular position vs time for the four different tested Voltages.

Table 1 displays the angular displacement and angular velocity of the encoder gear and the disk atop the centrifugal governor. The angular displacement and velocity of both components were measured at four different voltages to test the effectiveness of the mechanical rectifier.

The graphs in Table 1 show the angular velocity oscillating with large amplitudes according to the inputs of the encoder gear. The graphs of the angular velocity of the mechanical rectifier at these same voltages have a much smaller amplitude during the oscillation. This shows that the angular velocity of the output is extremely consistent compared to the input power. Furthermore, the graph of the rotation angle against time is linear, showing the consistency in movement of the output. Since the derivative of the rotation angle with respect to time is the angular velocity and the graphed angle is linear, the angular velocity would be constant. This is not the case as shown in the angular velocity graphs but it is very close to constant. The mechanical rectifier this group created severely dampened the amplitude of the oscillation, making the output essentially constant with negligible interference. This was shown by comparing the graphs of the angular velocity of the non-rectified system with the graphs of angular velocity of the rectified system. The large oscillations in the graphs of the former are severely dampened in the graphs of the latter. The linearity of the angle and the close-to-constant angular velocity proved that this system which used a centrifugal governor, a rubber band pulley system, and slippage between each subsystem was an effective mechanical rectifier.

$$dB = 20 * \log_{10}\left(\frac{A_1}{A_0}\right)$$

Voltage	Amplitude Without Rectifier(degrees/second)	Amplitude of Rectifier(degrees/second)	Decibel Scale
6V	171	16.9	-20.1
7.5V	237	25.9	-19.2
9V	259	17.7	-23.3
12V	337	31.6	-20.6

Table 2: Decibel scale calculations with amplitude.

The angular velocity of the non rectified system, regardless of voltage, oscillates with a much higher amplitude than the angular velocity of the rectified system. This is due to the level of dampening provided by the mechanical rectifier, which ranged from -23.3 dB to -19.2 dB. Even if the angular velocity vs. time graph for the rectified system still oscillates, its amplitude is so much lower than the unrectified system that it is relatively fixed compared to the unrectified system.

Future work & Conclusion:

If the group were to repeat this project in the future, several changes would have to be made to ensure a more functional result. For one, on a more logistical scale, the original design for the mechanical rectifier proved to be too difficult to execute, however the design was changed very late into the production timeline of the project. Thankfully, the original design and final design shared similar components, and as such not much work was scrapped. If this project were done again, the design would need to be fully formulated well before the production of parts to maximize efficiency.

Through the problems experienced during this project, the group learned valuable lessons in communication and working in a team. Redoing this project with that experience would cause much smoother and more efficient work. Furthermore, early failures with dimensioning would be solved with better understanding of what comes out of a 3D printer that was gained from failed prints. With more time, a more in depth analysis of the system and its flaws would be achieved and the group would work to improve these flaws. The group has issues finding the best way to transfer power. They decided on a rubber band pulley system due to the ease of use and the strength of the friction between the rubber band and the pulley. The belt was found to have a worse connection to the pulley and was difficult to assemble. While the rubber band was chosen, it wasn't the perfect choice. There were issues transferring power because the rubber bands have lots of tension in them and stretch. This caused imperfect power transfer because some power was lost in the rubber band. Furthermore, the rubber band tended to get caught and stop movement all together. In future iterations of this design a better method of power transfer would be found and tested. Another issue was the centrifugal governor wasn't perfectly connected to the encoder. Since the parts of the centrifugal governor that were connected to the pole were not friction fit, there was an imperfect power transfer between the encoder and the output. The group had concerns that this would negatively affect the final results. While testing, the group realized that this imperfect transfer of power was actually beneficial and the slippage between the centrifugal governor and the rest of the system was taking away energy that would have caused a larger oscillation in the output.

Through analysis of the oscillation of the output angular velocity, the group realized the output velocity was very close to constant and determined their mechanical rectifier a success. While they came across many difficulties that they will work to fix such as communication problems and failure in dimensioning and fitting parts and subsystems together, the group learned valuable skills in problem solving, the design process, power transfer, and teamwork.

Acknowledgements:

We would like to acknowledge Professor Micheal Giglia and Mr. Douglass Thornhill for their guidance throughout the project, as well as for posing this challenge in the first place.